5. Scenario: You are a data analyst working for a car manufacturing company. As part of your analysis, you have a dataset containing information about the fuel efficiency of different car models. The dataset is stored in a NumPy array named fuel_efficiency, where each element represents the fuel efficiency (in miles per gallon) of a specific car model. Your task is to calculate the average fuel efficiency and determine the percentage improvement in fuel efficiency between two car models.

AIM:

To calculate the average fuel efficiency and percentage improvement in fuel efficiency between two car models using NumPy arrays.

PROCEDURE:

1. Import the NumPy library.
2. Create a NumPy array containing fuel efficiency values.
3. Calculate the average fuel efficiency using np.mean().
4. Calculate the percentage improvement between two car models.
5. Display the average fuel efficiency and percentage improvement.

PROGRAM CODE:

```
import numpy as np

fuel_efficiency = np.array([25, 30, 35, 40])

average = np.mean(fuel_efficiency)

improvement = ((fuel_efficiency[3] - fuel_efficiency[0])

        / fuel_efficiency[0]) * 100
```

print("Average Fuel Efficiency:", average, "MPG")

print("Percentage Improvement:", improvement, "%")

INPUT:

Fuel efficiency = [25, 30, 35, 40] MPG

OUTPUT:

```
import numpy as np

fuel_efficiency = np.array([25, 30, 35, 40])

average = np.mean(fuel_efficiency)
improvement = ((fuel_efficiency[3] - fuel_efficiency[0])
               / fuel_efficiency[0]) * 100

print("Average Fuel Efficiency:", average, "MPG")
print("Percentage Improvement:", improvement, "%")
```

```
Average Fuel Efficiency: 32.5 MPG
Percentage Improvement: 60.0 %
```

RESULT:

Thus, the average fuel efficiency is 32.5 MPG, and the percentage improvement between the first and fourth car models is 60%.